

Carlos De Manuel Vicente

Data & Analytics Engineer

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Data & Analytics Engineer at a global automotive manufacturer. I build the models, pipelines and apps that 20+ plants in North America, APAC and EMEA use every day, from the reports top management reads to the tools people use on the shop floor. Stack: Snowflake, dbt, Python, SQL, Prefect, AWS, Power BI and Streamlit.

EXPERIENCE

Data & Analytics Engineer Jan 2025 – Present
RESRG Automotive · global automotive manufacturer Liria, Spain (hybrid)

- Replaced manual Excel trackers across **20+ plants** (NA, APAC, EMEA) with self-serve data products in Streamlit and Power BI, from the executive summary **200+ people** open daily to inventory apps on the shop floor, across purchasing, finance, supply chain and production.
- Built near-real-time analytical models for a warehouse management system over **800M+ rows** in Snowflake, refreshed by event-driven Prefect flows that fire when upstream Fivetran HVR replication lands, with the data engineering team.
- Set one definition for **130+ operational KPIs** through a KPI-governance app built on dbt, with tests on every model and pipeline observability in Prometheus.
- Re-engineered the end-to-end refresh with CI/CD, cutting runtime **~98%** and compute cost **~90%**.
- Forecast plant electricity consumption from production variables for the purchasing team, joining utility invoices with production and financial data so usage normalises against sales.

Data Scientist (internship) Nov 2023 – Jul 2024
Biomedical Data Science Lab, ITACA, UPV Valencia, Spain

- Built an end-to-end Python framework and preprocessing pipelines to evaluate healthcare ML models for fairness, robustness and explainability (XAI).

PROJECTS

Job Market Intelligence · [case study](#) · [github.com/carlosdmv7/job-market-intelligence](#)
Reads EU data-job ads every day and ranks them against a CV: six free APIs, LLM enrichment that went from **23 to 180** ads a day on the same free quota, dbt on MotherDuck, Prefect Cloud and Streamlit.

Spanish Housing Radar · [case study](#) · [github.com/carlosdmv7/spanish-housing-radar](#)
Finds under-priced flats in València by scoring every listing against its neighbourhood: weekly scraping, dbt with **162 data tests**, Prefect Cloud, Docker and Streamlit.

SKILLS

Warehouse & modelling	Snowflake, dbt, DuckDB, Kimball dimensional modelling
Languages	Python, SQL, Jinja, DAX
Platform & orchestration	AWS, Prefect, Airflow, GitHub Actions (CI/CD), Git, Azure DevOps, Docker
Quality & observability	pytest, Pydantic, ruff, mypy, Prometheus
BI & data apps	Power BI, Streamlit, Microsoft Fabric
AI & ML	Gemini and OpenAI APIs (incl. Azure OpenAI), LLM evals, scikit-learn, SHAP

EDUCATION

MSc, Data Analysis, Process Improvement & Decision Support Engineering · UPV, Valencia

BSc, Biomedical Engineering (ICT specialisation) · UPV, Valencia

Thesis graded 10/10 · distinction for academic performance (top 25%)

LANGUAGES

Spanish (native) · English (C1, Cambridge, 194)